

Financial Analysis Report of The Boeing Company

Karthik Krishnagiri Pachiyappan
University of Massachusetts Amherst
M&IE-ENG 645 Project Budgeting & Finance
Prof. Doug Eddy
March 14, 2025

Introduction

The Boeing Company is a leading aerospace manufacturer known for its commercial aircraft, defense systems, and space technology. As a major player in the global aviation industry, Boeing's financial health is of critical importance to investors, stakeholders, and policymakers. This report examines Boeing's financial position by analyzing key financial ratios, including profitability, solvency, liquidity, and operational efficiency. Additionally, the report addresses key business questions regarding Boeing's market position, competitive advantages, and financial outlook based on its latest financial statements and disclosures.

Company Overview

Founded in 1916, The Boeing Company is a global leader in aerospace and defense, headquartered in Arlington, Virginia. It designs, manufactures, and services commercial airplanes, military aircraft, defense systems, and space technology. With operations spanning over 150 countries, Boeing serves airlines, defense forces, and space agencies worldwide.

Operating in a capital-intensive industry, Boeing invests heavily in research and innovation to stay competitive. It faces strong competition from Airbus in commercial aviation and Lockheed Martin in defense. The company generates revenue through aircraft sales and services, including maintenance and support. Despite financial hurdles in recent years, Boeing remains a key player in shaping the future of aviation, defense, and space exploration through advanced technology and engineering expertise.

Business and Market Analysis

Business Categorization

Boeing is best known for building commercial and military aircraft, satellites, and defense systems, making it primarily a product-based company. However, it also plays a key role in aviation services, offering maintenance, training, and support through its Global Services segment. This combination of products and services strengthens customer relationships and provides a steady revenue stream. While its Commercial Airplanes division drives most of its earnings, the services segment remains a vital part of its financial stability.

Competitive Preference

Boeing stands out in the aerospace industry due to its technological advancements, strong reputation, and extensive supply chain. It invests heavily in research and development to improve aircraft efficiency, avionics, and overall performance. Its long-standing relationships with airlines and governments, especially through defense contracts which provide financial security and reinforce its leadership in the market.

Commodity-Based Business and Capital Intensity

Boeing operates in an industry that demands huge investments in manufacturing, materials, and regulatory compliance. While some aspects of aircraft manufacturing resemble commodity-based businesses such as pricing pressures and market cycles, Boeing differentiates itself through innovation and branding. Strict certification requirements and government regulations make it difficult for new competitors to enter the market, securing Boeing's position as an industry leader.

Footnote Disclosures

Boeing's financial statements offer important insights into its business. The footnotes cover areas like accounting policies, pension obligations, long-term contracts, and legal risks. They also highlight production costs, revenue recognition methods, and contingent liabilities. Pension obligations remain a major financial commitment, while details on segment performance provide a deeper understanding of Boeing's overall efficiency.

Business Outlook

Boeing's future looks cautiously optimistic, with growth opportunities tempered by operational challenges. The rising demand for fuel-efficient aircraft and expansion into space exploration create strong potential, but ongoing supply chain issues and regulatory hurdles pose risks. The company is focused on stabilizing cash flow, but production delays remain a key concern for financial performance.

Industry Comparisons

Boeing competes with major industry players like Airbus, Lockheed Martin, and Northrop Grumman. While Airbus currently holds a larger market share in commercial aviation and has shown stronger profitability in recent years, Boeing continues to secure a high volume of aircraft

orders, demonstrating strong demand for its products. These industry comparisons help investors and analysts assess Boeing's strengths and areas where it needs to improve.

Financial Analysis

Profitability Analysis

Profitability ratios provide a company's ability to earn profits from its sales, balance sheet assets, or shareholders' equity. They indicate how efficiently a company generates profit and value for shareholders.

1. **Return on Assets (ROA):** Boeing's ROA was positive from 2015 to 2018, peaking at 9.11% in 2018. However, it declined significantly after 2019, reaching -8.31% in 2024. This downward trend indicates declining efficiency in using assets to generate profits, mainly due to increased costs and reduced revenues. (Chart 1)
2. **Return on Investment (ROI):** Boeing's ROI showed strong performance before 2018 but turned negative in 2019 and continued to decline, reaching -29.88% in 2024. The negative ROI highlights inefficient capital utilization and financial struggles. (Chart 2)
3. **Return on Equity (ROE):** ROE was erratic, peaking at 25.49% in 2018 but turning negative from 2019 onward. A negative ROE indicates that Boeing's losses have eroded shareholder value. (Chart 3)

Solvency Analysis

Solvency or leverage ratios measure the extent to which a company uses debt to finance its operations and the company's ability to meet its financial obligations.

1. **Debt Ratio:** Boeing's debt ratio increased significantly after 2019, peaking at 0.41 in 2020 and 2021 before stabilizing at 0.34 in 2024. This suggests a high reliance on debt to finance operations. (Chart 4)
2. **Debt-to-Equity Ratio:** The debt-to-equity ratio turned negative after 2019, indicating negative shareholder equity due to accumulated losses. This signals financial distress and increased dependence on borrowed capital. (Chart 5)

Liquidity Analysis

Liquidity ratios measure the ability of a company to pay off its short-term liabilities.

1. **Current Ratio:** Boeing's current ratio fluctuated but remained above 1.0, indicating its ability to meet short-term obligations. The ratio improved to 1.39 in 2020 but declined in subsequent years. (Chart 6)
2. **Quick Ratio:** The quick ratio remained below 0.5, signifying liquidity challenges. A low quick ratio indicates that Boeing has limited liquid assets to cover short-term liabilities without relying on inventory sales. (Chart 7)

Efficiency Analysis

Activity ratios or efficiency ratios measure how effectively a company utilizes its assets to generate revenue or cash flow.

1. **Inventory Turnover Ratio:** Boeing's inventory turnover ratio fluctuated, dropping significantly after 2019. A declining ratio suggests slower inventory movement, indicating potential inefficiencies in production and sales. (Chart 8)
2. **Days of Sales Outstanding (DSO):** DSO increased significantly after 2018, peaking at 66.67 days in 2021. Higher DSO implies delayed collection of receivables, negatively affecting cash flow. (Chart 9)

Conclusion

Boeing remains a dominant force in the aerospace and defense industry, but its financial performance in recent years highlights both strengths and vulnerabilities. The company's diversified revenue streams, strong market position, and technological advancements provide a solid foundation for future growth. However, financial indicators such as declining profitability, increased debt reliance, and liquidity challenges suggest ongoing struggles.

While Boeing continues to secure significant aircraft orders and defense contracts, operational inefficiencies and external pressures, such as supply chain disruptions and regulatory scrutiny, pose risks. The company's cautious optimism is warranted, as future success will depend on stabilizing financial performance, addressing production delays, and capitalizing on emerging opportunities in commercial aviation and space exploration.

References

1. The Boeing Company Financial Statements (10-K)

Year	Financial Report Links
2024	Financial Report 2024
2023	Financial Report 2023
2022	Financial Report 2022
2021	Financial Report 2021
2020	Financial Report 2020
2019	Financial Report 2019
2018	Financial Report 2018
2017	Financial Report 2017
2016	Financial Report 2016
2015	Financial Report 2015

2. Drummond, C. K. (2018). *Financial decision-making for engineers*. Yale University Press
3. [Boeing Company Reports](#)
4. [Macro Trends Reports](#)
5. [Excel workbook for Data and Charts](#)

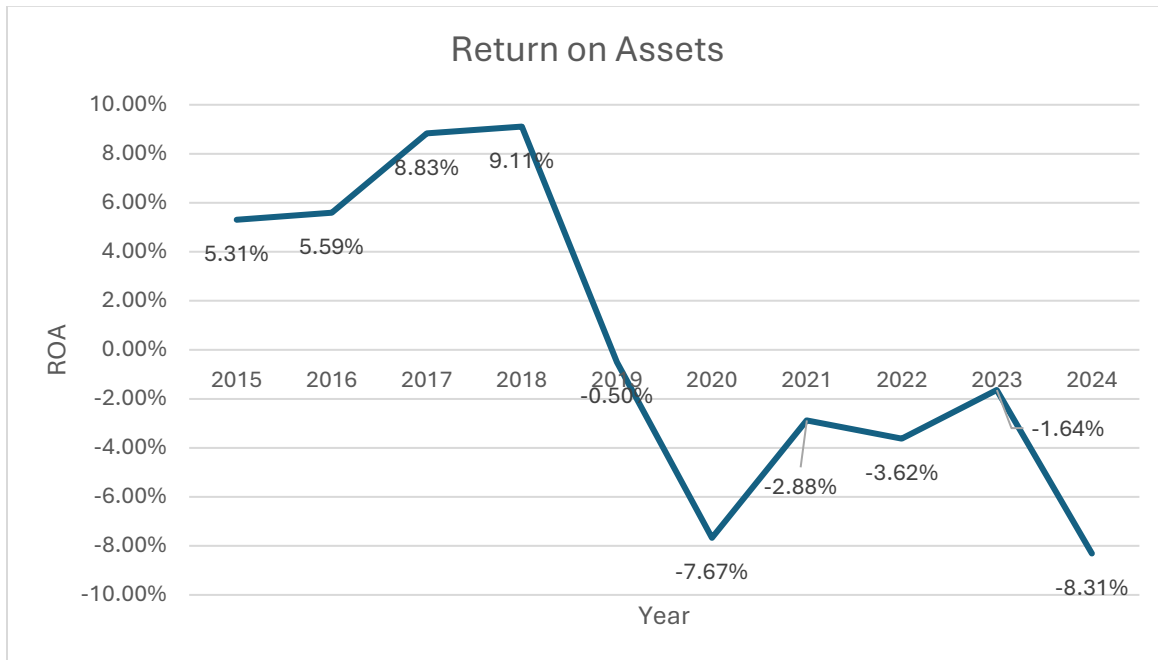
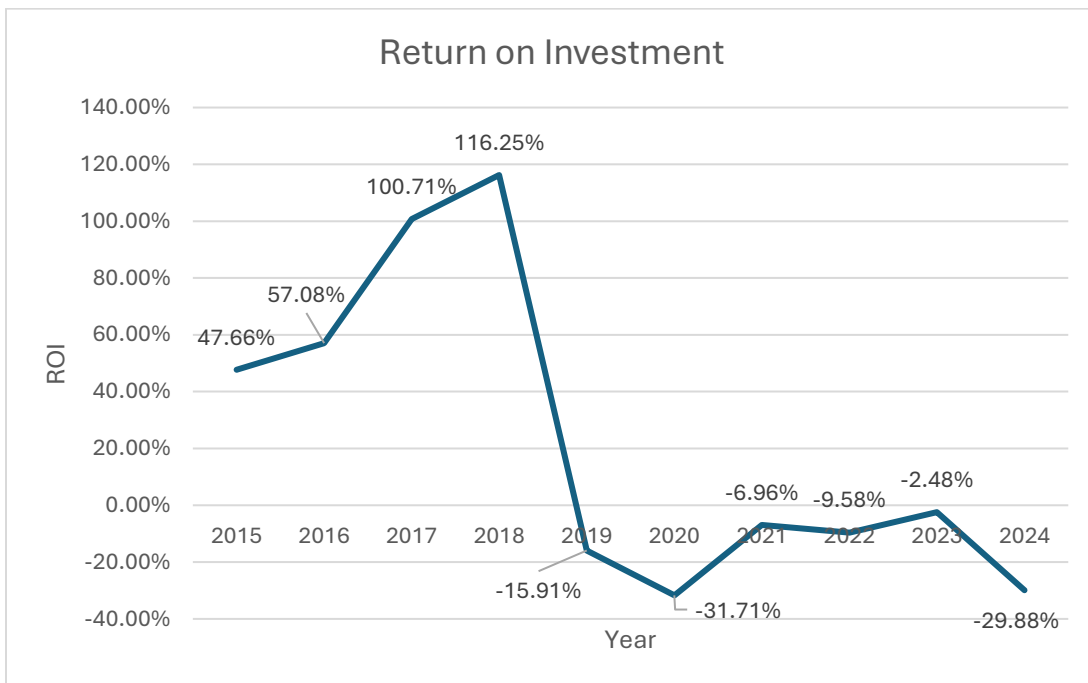
Appendix**Chart 1 - Return on Assets (ROA)****Chart 2 - Return on Investment (ROI)**

Chart 3 - Return on Equity (ROE)

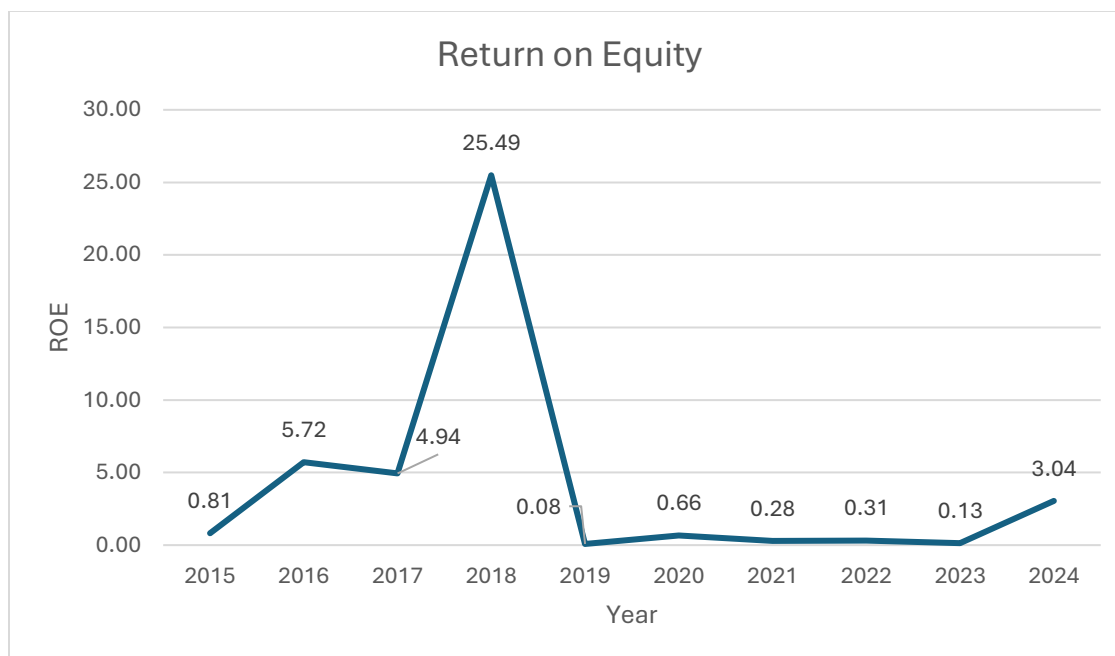


Chart 4 - Debt Ratio

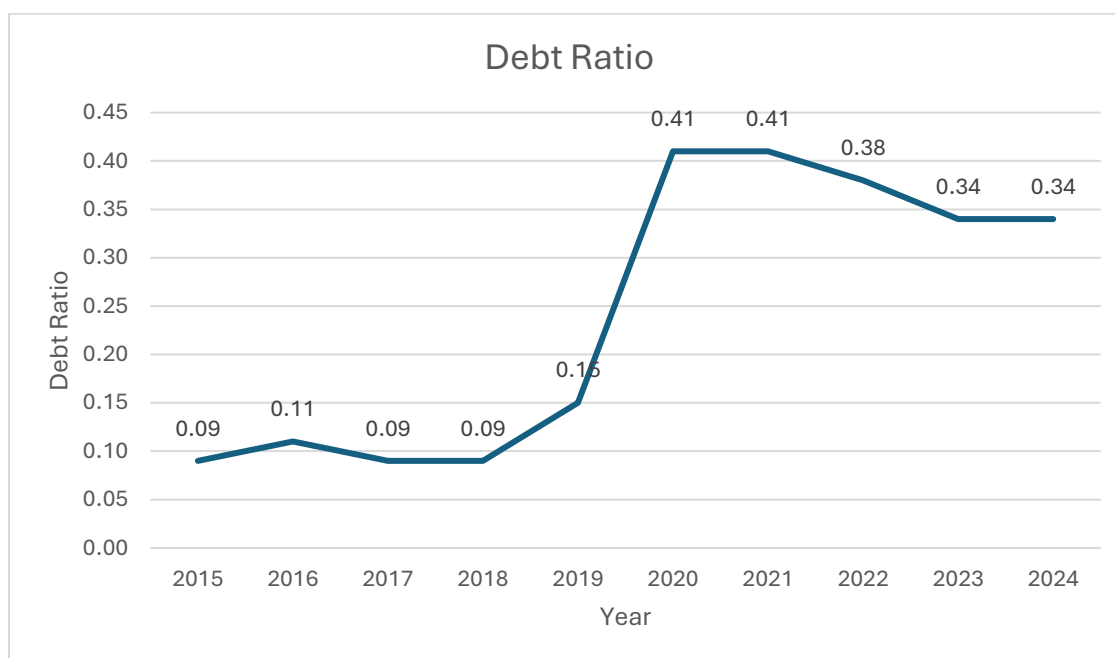


Chart 5 - Debt-to-Equity Ratio

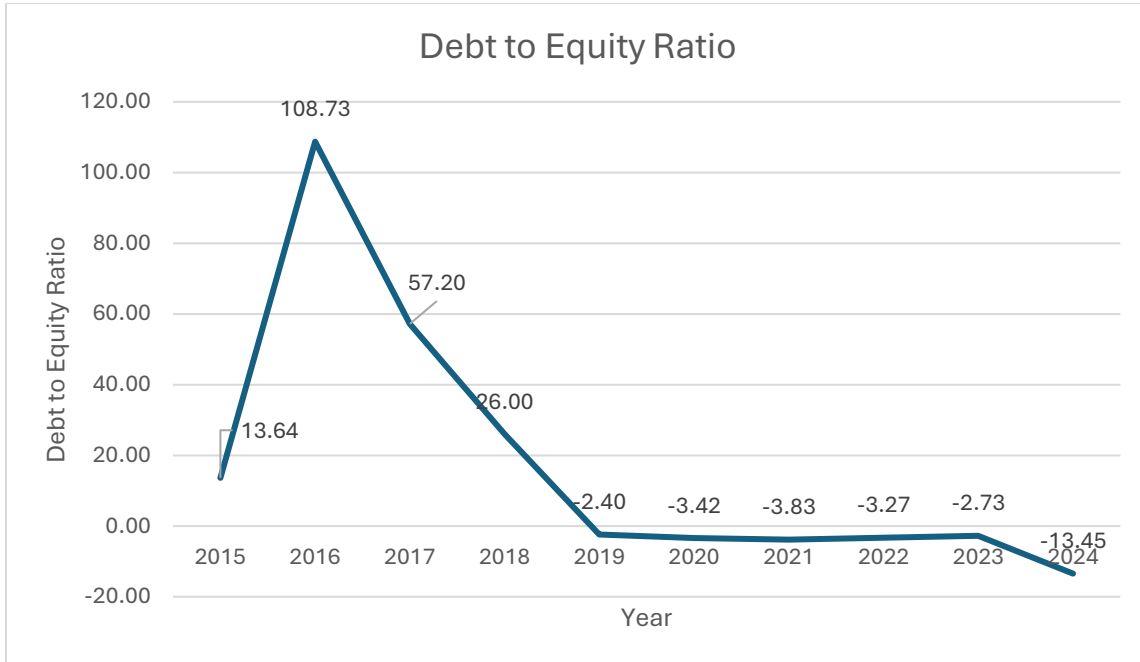


Chart 6 - Current Ratio

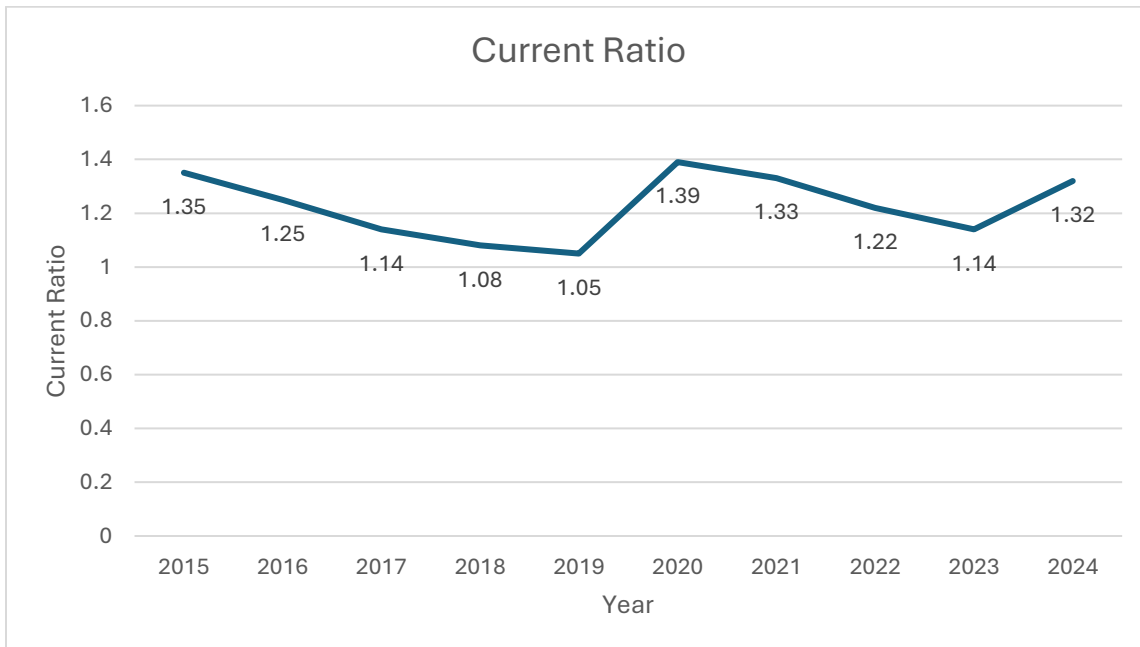


Chart 7 - Quick Ratio

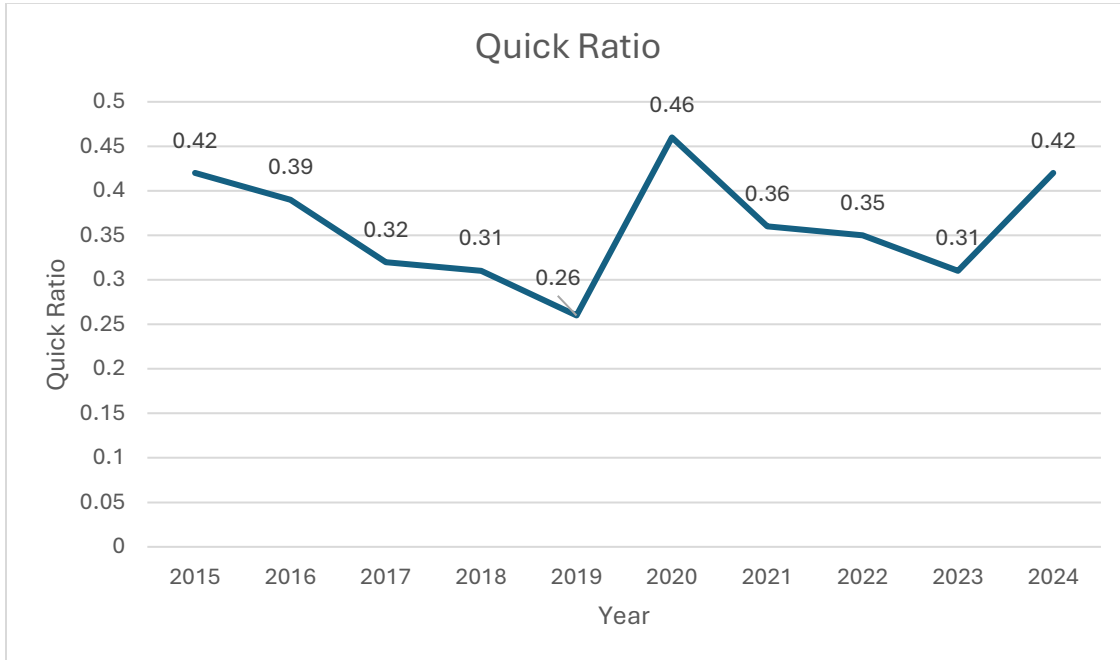


Chart 8 - Inventory Turnover Ratio

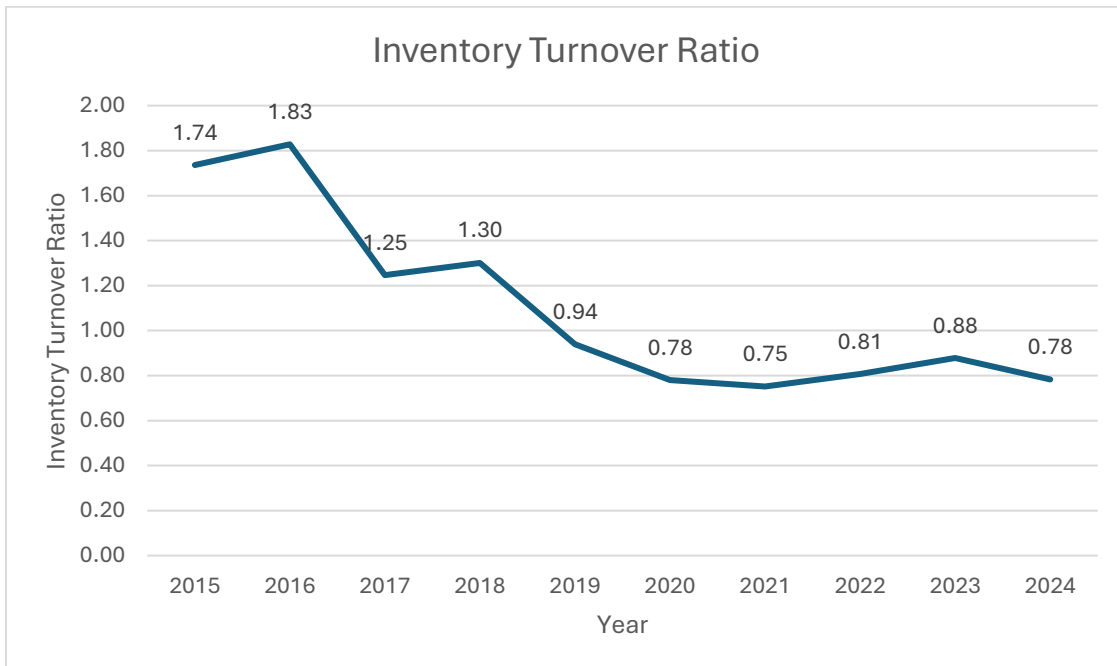


Chart 9 - Days of Sales Outstanding (DSO)

